

MODULE 05

Hip MRI & THA: AVN Staging, FAI & Arthroplasty Complications

Ficat-Arlet · ARCO · Alpha Angle · LCEA · FAI · Vancouver · ALTR · Brooker

1.0 AMA PRA Cat 1

Hip

CME Article Review

LEARNING OBJECTIVES

1. Apply Ficat-Arlet and ARCO classifications to femoral head AVN on MRI and radiograph.
2. Calculate and interpret alpha angle, LCEA, and femoral anteversion on hip MRI.
3. Classify FAI subtypes (cam, pincer, mixed) and their MRI features.
4. Identify and grade THA complications: ALTR (Anderson), periprosthetic fracture (Vancouver).
5. Apply Brooker classification to heterotopic ossification post-THA.
6. Recognize the crescent sign as the critical juncture for joint-preserving surgery.

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1. AVN — Ficat-Arlet & ARCO Staging

Ficat-Arlet Classification

Stage	Radiograph	MRI Finding	Management
I	Normal	Double line sign on T2; T1 band hypointensity	Core decompression ± BM concentrate
IIA	Sclerosis or cysts; no crescent	Established signal change	Core decompression ± grafting
IIB	CRESCENT SIGN — subchondral Fx	T2 fluid beneath articular cortex	Core decompression; TPVIF; impending collapse
III	Femoral head COLLAPSE; acetabulum normal	Flattening; collapse evident	Resurfacing (young); THA (older)
IV	Collapse + GH arthrosis	Arthrosis changes	THA required

CRITICAL

The CRESCENT SIGN = ARCO Stage 3 = subchondral fracture = LAST CHANCE for joint-preserving surgery. Beyond this, collapse is inevitable without intervention. Report presence or absence explicitly.

2. FAI — Hip MRI Measurements

FAI Subtypes

Type	Morphology	Key Measurement	Cartilage Damage Pattern
Cam	Aspherical head; pistol-grip; reduced offset	Alpha angle >55°	Superior/anterosuperior acetabular cartilage (outside-in)
Pincer	Acetabular overcoverage; retroversion	LCEA >40°; crossover sign	Posterior labrum; contrecoup damage
Mixed	Both cam + pincer features	Combined findings	Combined; most common type

Key Hip MRI Measurements

Measurement	Normal	Threshold	Significance
Alpha angle	<55°	>55° = cam morphology	Head-neck asphericity; impingement
LCEA (Wiberg)	25–40°	<25° = dysplasia; >40° = overcoverage	Lateral acetabular coverage

Measurement	Normal	Threshold	Significance
Femoral anteversion	10–25°	<10° or >35° = abnormal	Rotational profile; instability
Tönnis angle	<10°	>10° = undercoverage	Roof obliquity; dysplasia

3. THA Complications — Anderson ALTR, Vancouver & Brooker

Anderson ALTR Classification (MARS MRI)

Type	Description	Management
1 — ALVAL	Synovial thickening; fluid; no mass	Monitor; interval MRI
2A — Solid	Solid periarticular mass	Surgical if symptomatic/enlarging
2B — Cystic	Cystic/complex fluid collection; thin wall	Surgical if large/symptomatic
3 — Mixed	Solid + cystic; aggressive features	Early revision recommended
4 — Destructive	Bone or muscle destruction; involvement	Urgent revision

Vancouver Periprosthetic Femoral Fracture

Type	Location	Stem Stability	Strategy
AG	Greater trochanter	N/A	ORIF if >2.5 cm displaced
AL	Lesser trochanter / calcar	Usually stable	Conservative if <2 cm + stable stem
B1	At or around stem tip	STABLE	ORIF with plate + cerclage; retain stem
B2	At or around stem tip	UNSTABLE (loose)	Revision to long-stem prosthesis
B3	At stem tip + poor bone	Unstable + deficit	Revision + allograft or proximal fem replacement
C	DISTAL to stem tip	N/A (uninvolved)	ORIF; retain stem regardless

Brooker Heterotopic Ossification

Grade	Finding	Impact
I	Islands of bone in soft tissues	Minimal
II	Bone spurs; ≥1 cm gap	Mild; monitor
III	Bone spurs; <1 cm gap	ROM restriction
IV	Bony ankylosis	Severe; delayed excision

- Ficat RP. J Bone Joint Surg Br 1985.
- Anderson H et al. J Arthroplasty 2017.
- Duncan CP, Masri BA. Instr Course Lect 1995.
- Brooker AF et al. J Bone Joint Surg Am 1973.